MIL-C-43843A
20 May 1985
SUPERSEDING
MIL-C-43843
30 March 1973

MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, POLYESTER/COTTON, PRECURED, DURABLE PRESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers blended polyester and cotton cloth, treated with a cured durable press finish.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon Covered

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Method 124 - Appearance of Durable Press Fabrics after Repeated Home Launderings

(Copies of Test Method may be obtained upon application to American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard sample. The dyed, treated, and cured cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials.

3.2.1 Fibers.

3.2.1.1 Polyester. The fiber shall be polyethylene glycol terephthalate.

3.2.1.2 Cotton. The cotton shall be carded or combed.

3.2.2 Yarn. The warp and filling yarn shall be singles made from a blend of 65 ± 3 percent polyester and the remaining percentage cotton, based on the dry weight of the desized cloth when tested as specified in 4.2.3.

3.3 Color. Unless otherwise specified (see 6.2), the color of the dyed, treated and cured cloth shall be Black 305 and shall match the standard sample (see 6.3).

3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.2.3.

3.3.2 Matching. The dyed, treated, and cured cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 100 K.

3.3.3 Colorfastness. The dyed, treated, and cured cloth shall show fastness to light, laundering (after 3 cycles), perspiration, and crocking equal to or better than the standard sample. When no standard sample is

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available, the dyed, treated, and cured cloth shall show "good" fastness to light, laundering (after 3 cycles), perspiration, and crocking when tested as specified in 4.2.3.

3.4 Physical requirements. The dyed, treated and cured cloth shall conform to the requirements specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Weight ounces per sq yd minimum	Yarns per inch minimum		Breaking strength pounds, minimum	
	Warp	Filling	Warp	Filling
4.6	102	56	125	90

3 4.1 Weave. The weave shall be a plain weave.

3.4.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages when fly shuttle looms are used, and exclusive of selvages and fringe when shuttleless looms are used.

3.5 Cloth preparation. The cloth shall be desized, scoured, mercerized, bleached, and heat set prior to dyeing.

3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the dyed cloth prior to treatment shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.7 Dimensional stability. The dyed, treated, and cured cloth shall not shrink or elongate more than 2.0 percent in either the warp or filling direction when tested as specified in 4.2.3.

3.8 Finish. The cloth shall be given an approved durable press treatment (see 6.5).

3.9 Appearance rating. The dyed, treated, and cured cloth shall have an appearance rating of not less than 3.5 when tested after five launderings as specified in 4.2.3.

3.10 Seam efficiency. The dyed, treated, and cured cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.2.3.

3.11 pH. The pH value of the water extract of the cured cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

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3.12 Length and put-up. Unless otherwise specified (see 6.2), the cured cloth shall be furnished in continuous lengths each not less than 40 yards. The cloth shall be put-up on rolls as specified in 5.1.

3.13 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.14 Workmanship. The dyed, treated, and cured cloth shall conform to the quality established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise speci-fied in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the

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limits shown on Sears Fabric Defect Scale (see 6.4), E or 3 as applicable for slubs and B for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square of the total yardage examined exceeds 30.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 45.0 points. If one roll exceeds 45.0 points per 100 square yards, a second sample of the size indicated above shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension - one point

For defects exceeding 3 inches but not exceeding 6 inches in any dimension - two points

For defects exceeding 6 inches but not exceeding 9 inches in any dimension - three points

For defects exceeding 9 inches in any dimension - four points

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur:

Holes, cuts, or tears
Baggy, ridgy, or wavy cloth
Width less than minimum specified.
Poor dye penetration, mottled, streaky, or cloudy
Overall uncleanness

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4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll off in the sample off shade; shaded side to side, side to center, or end to end; or not having the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average result of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values used in expressing the final results. The sample unit shall be 4 continuous yards full width of the finished cloth and 1/4 yard of the cloth prior to application of the durable press treatment. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Polyester identification	3.2.1.1	1600 1/
Cotton identification	3.2.1.2	1200 1/
Fiber content	3.2.2	2535 1/
Yarn (single)	3.2.2	Visual 1/
Labile sulfur	3.3.1	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.3.3	5610 2/
Crocking	3.3.3	5651
Perspiration	3.3.3	5680
Light	3.3.3	5660 3/
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Weave	3.4.1	Visual 4/
Mercerization	3.5	1/
Nonfibrous material	3.6	2611
Dimensional stability	3.7	4.3.1
Approved treatment	3.8	1/
Appearance rating	3.9	4.3.1
Seam efficiency	3.10	5110 5/
pH	3.11	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens shall be dried after each of the three launderings cycles.

3/ The exposure time shall be 80 hours.

4/ One determination per sample unit and the result reported as "pass" or "fail".

5/ The needle shall measure 0.36 inch ± 0.001 inch across the blade at the eye. The polyester/cotton wrapped thread shall conform to MIL-T-43548. The needle thread shall be 70, 2 ply and the looper thread shall be 80, 2 ply.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

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4.3 Methods of inspection.

4.3.1 Appearance and dimensional stability after laundering testing.

4.3.1.1 Apparatus and materials. Apparatus and materials shall be as specified in AATCC method 124.

4.3.1.2 Preparation of specimens. Three specimens 22 by 22 inches measured parallel to the warp and filling shall be cut from a portion of the cloth sample. One specimen shall be cut from each side of the sample unit to within 3 inches of the selvages and the third specimen shall be taken from the center. No two specimens shall contain the same filling yarns. The specimens shall be conditioned to equilibrium under Standard Conditions in accordance with FED-STD-191. The conditioned specimens shall be laid without tension on a flat surface, care being taken that the cloth is free from wrinkles or creases. Three distances, each a minimum of 18 inches, shall be measured and marked off parallel to each of the warp and filling directions of the specimen. Each pair of markings shall be a minimum of 6 inches from each other and not closer than 1 inch to the edges of the specimen. The distance may be marked with indelible ink and a fine pointed pen, or by sewing fine threads into the cloth, or by stamping. The samples shall then be laundered in accordance with 4.3.1.3.

4.3.1.3 Laundering. Place the three 22 by 22 inch specimens in the washer. The washing load shall be 4 $\pm$ 1/4 pounds. Dummy pieces shall be added to the machine, along with the specimens to make up the 4 $\pm$ 1/4 pound load. Fill to the full water level of the washer with water of a hardness not to exceed 50 parts per million and at a temperature of 120° $\pm$ 5°F. The rinse temperature shall be 80° $\pm$ 5°F. Add 140 $\pm$ 1 grams of detergent. Set the washer for a 12-minute cycle on the "Permanent Press" setting. Allow the washing to proceed automatically through the final spin cycle. Remove the specimens immediately at the completion of the final spin and separate from the dummy pieces and each other if tangled. Place the complete washed load (4 $\pm$ 1/4 pounds) in the dryer and dry at the "Permanent Press" setting with a 10-minute cool down cycle. Operate the dryer until the load is dry and continue tumbling 5 minutes with the heat turned off (cool down cycle). Remove the load immediately after the machine stops. Repeat the wash and dry cycles for five complete cycles. Remove all the specimens and condition to equilibrium under standard conditions in accordance with FED-STD-191.

4.3.1.4 Appearance evaluation. Three trained observers shall evaluate each sample unit for appearance characteristics. The observers shall make their evaluations independent of each other. Each observer shall evaluate each specimen for appearance as it hangs on the viewing board while standing in front of the viewing board and 4 feet back from it. The overhead lighting shall be used. Mount each 22 by 22 inch specimen on the viewing board with the center of the specimen 5 feet from the floor. Place three dimensional

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plastic replicas on each side of the specimen with the centers 5 feet from the floor to facilitate comparative rating. The specimen shall be rated according to the appearance on the plastic replica that most nearly matches the appearance of the cloth. No estimated rating falling between the approved replica shall be given. The average of the rating values assigned for appearance of the cloth by the observers rounded off to the nearest 0.1 rating shall be the rating for the specimen. Each rating value given by the observers for the appearance of the cloth shall also be reported.

4.3.1.5 Dimensional stability. After evaluation for appearance is made, the three 22 by 22 inch specimens shall be laid out without tension on a flat surface in the standard atmosphere. Care shall be taken that the specimens are free from wrinkles and creases. The previously measured distance marked on the specimens shall be measured in both the warp and filling directions. The dimensional stability of the specimens shall be calculated as follows:

$$\text{Dimensional change, percent} = \frac{A-B}{A} \times 100$$

Where A = Average of initial measurements (3 specimens)
B = Average measurements after laundering (3 specimens)

The dimensional change of the sample unit in the warp and filling directions shall be the average of the specimens tested in each direction respectively and shall be reported separately to the nearest 0.1 percent. The individual values used to calculate the average shall also be reported.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth, shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required in the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

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6.1 Intended use. The cloth is intended for use for necktabs in the women's polyester/cotton shirt.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Color of cloth required if other than Black 305 (see 3.3).
c. Width of cloth required (see 3.4.2).
d. Length of roll required if other than specified (see 3.12).
e. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids (see 3.1).

6.4 Fabric defect scales. Fabric Defect Replica Kits are available from Sears Roebuck and Company, Department 817 (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

6.5 Durable press resin treatment. Approval of the durable resin for utilization under this document is the responsibility of the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity which are not set forth in this document. Because of the time necessary to conduct full evaluation (approximately 6 months) only those treatments approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement.

6.6 Yarn number. The cloth has been satisfactorily manufactured using the following cotton count yarn numbers:

27/1 warp and 22.5/1 filling

6.7 Changes from previous issue. Asterisks are not use in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0014

Review activities:

Army - MD
Navy - MC
Air Force - 82, 99
DLA - CT

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US Army Natick Research and Development Center
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-43843A	Cloth, Plain Weave, Polyester/Cotton, Precured, Durable Press

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK/OT, 1 Apr 84

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